Midterm Study Sheet

EE5203 Microprocessors — what the exam covers (L01–L07), how the questions look, and practice with answers

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Rule: try every practice item on paper before you open the answers at the end. The exam is written by hand, closed book, 80 minutes. Practice the same way: no compiler, no debugger, a pen and this sheet.

How the exam works

Section	Questions	Points	What you do
A — Trace the execution	4 × 6p	24	Read a short C program. Write exactly what it prints, or the final register / variable values.
B — Multiple choice	5 × 4p	20	One correct answer per question. Concepts from L01–L07.
C — Fill in the blank	2 × 5p	10	A working program with five blanks. One point per blank.
D — Short code	4 questions	46	Write a small function or a loop body. Minimal C; comments welcome.

Every question gives you the facts it needs: the pin, the bit positions, the timer clock, the HAL names. The first page of the exam is a reference box with the HAL and CMSIS names you may use. You do **not** memorise register addresses, bit encodings or function names. You do need to be fluent in the patterns below.

Board facts you may assume: Nucleo-F401RE, LD2 on PA5, B1 on PC13 (external pull-up, pressed = pin low), timer clock 16 MHz.

L01 — Toolchain and evidence

Stage	Tool	Its message when it fails	Typical cause
Compile	compiler	`error: ... undeclared`, `expected ';'`	typo, missing declaration, syntax
Link	linker	`undefined reference to 'f'`	`f` is declared but **defined nowhere** in the build (the `.c` file is not in the build, or the body was never written)
Flash	ST-LINK	connection / target errors	cable, board, wrong project
Run	the board	nothing — you must observe	logic or configuration error

- A clean build proves the **source is valid C**, not that the board does what you want.
- A successful flash proves the firmware **reached the board**, not that it behaves correctly.
- Evidence is an observation: the LED, a variable in the debugger, a register in the SFR view.
- `HAL_Delay(100)` blinks faster than `HAL_Delay(500)` because the loop waits less between toggles.

L02 — Values, loops, decisions, arrays

Rule	Example
Integer division discards the fraction	`17 / 5` is 3; `17 % 5` is 2
`=` stores, `==` compares	`if (x = 1)` is a bug: it stores 1 and is always true
`>` is strict, `>=` is not	`3000 > 3000` is false; `3000 >= 3000` is true
`for (int i = 0; i < N; i++)` runs the body `N` times	`i` is `0` … `N-1` inside; `N` is never used as an index
Valid indices of `uint16_t a[N]`	`0` … `N-1`; `a[N]` is outside the array
Accumulators start at a known value	`sum = 0`, `count = 0`, `maximum = a[0]`
Clamp before you add	`if (a[i] > 4095) a[i] = 4095;` then `sum += a[i];`

How to trace a loop on paper: make a table with one row per repetition and one column per variable. Write the value of every variable **after** the body. Stop when the condition is false. Then compute what is printed.

`printf` in the trace questions. We hardly used `printf` in the labs (the board has no screen); in the exam it is only the way a question asks "what are the values now?". `printf("format", a, b, …)` prints the format text and replaces each `%…` with the next value, in order. Everything else in the format text is printed as it is.

Specifier	Prints	Example
`%d`	a signed integer (`int`)	`printf("%d", -5)` → `-5`
`%u`	an unsigned integer (`uint32_t`, `uint16_t`, `uint8_t`)	`printf("%u", 14995U)` → `14995`
`%02X`	the value in hexadecimal, capital letters, at least 2 digits, zero-padded	`0x4C` → `4C`, `0xC` → `0C`
`%08X`	the same with 8 digits	`0x40C` → `0000040C`
`\n`	ends the line	

- `printf("%d %d\n", 12, 13)` prints `12 13`: one space between the numbers because there is one space in the format text. `"%d,%d"` would print `12,13`.
- An expression is evaluated first: `printf("%d", sum / 6)` prints the result of the integer division, not the expression.
- On the answer line write exactly what is printed: the digits and spaces, no quotes, no units, no variable names.

L03 — Functions, arrays as parameters, strings

- A declaration (in the `.h`) has three parts: **return type, name, parameters**. The definition (in the `.c`) adds the body. The call uses the return value.
- **Parameters are copies.** Changing a parameter never changes the caller's variable. To send a result back, `return` it, and the caller must **store** it: `x = f(x);`.
- An array parameter `uint16_t values[]` is **not a copy** and does not know its own length: the caller passes `count`. Loop with `i < count`.
- Sum many `uint16_t` values into a `uint32_t`: 17×4095 already exceeds 65535.
- `#define SAMPLE_COUNT 8` is text replacement; it is not a variable.
- A C string ends with `'\0'`. `"HIGH"` needs **5** bytes. `{'O','K'}` is not a string. Compare strings with `strcmp(a, b) == 0`, never with `==`.

L04 — Pointers, structures, bits, registers

```
int x = 5;      int *p = &x;      /* p holds the ADDRESS of x           */
*p = 9;                            /* write THROUGH p: x is now 9        */
s.member                           /* member of a structure value        */
p->member   /* same as (*p).member: member through a pointer             */
```

Mask recipes (`n` = bit number, `k` = first bit of a 2-bit field = 2 × pin):

```
r |=  (1U << n);                      /* set bit n, other bits unchanged      */
r &= ~(1U << n);                      /* clear bit n: ~ inverts the mask      */
r ^=  (1U << n);                      /* toggle bit n                         */
if (r & (1U << n)) { /* bit n is 1 */ } /* test bit n: non-zero means set      */
r &= ~(3U << k);  r |= (pattern << k); /* replace a 2-bit field: CLEAR FIRST   */
```

Hex binary: one hex digit = four bits. `0x3 = 0011`, `0xC = 1100`, `0xF = 1111`. `1U << n` is the hex number with a single 1 at bit n: `1U << 4 = 0x10`, `1U << 12 = 0x1000`, `3U << 12 = 0x3000`.

GPIO registers (CMSIS names):

Register	Rule
`RCC->AHB1ENR`	bit 0 = GPIOA clock, bit 1 = GPIOB, bit 2 = GPIOC. **Enable the clock first**; an unclocked port ignores writes.
`GPIOx->MODER`	pin n owns bits 2n+1:2n. `00` input, `01` output, `10` alternate, `11` analog. Read-modify-write.
`GPIOx->BSRR`	bit n (0–15) sets pin n **high**; bit n+16 sets it **low**. Plain `=` is correct: each write is a command.
`GPIOx->IDR`	input level of all 16 pins; `(IDR >> n) & 1U` reads pin n.
`GPIOx->ODR`	current output levels; `^=` toggles a pin.

```
/* worked example: an external LED on PB7 */
RCC->AHB1ENR |= (1U << 1);      /* GPIOB clock (bit 1)                 */
GPIOB->MODER &= ~(3U << 14);    /* clear PB7 field (bits 15:14 = 2·7)  */
GPIOB->MODER |=  (1U << 14);    /* 01 = output                         */
GPIOB->BSRR   =  (1U << 7);     /* PB7 high                            */
GPIOB->BSRR   =  (1U << 23);    /* PB7 low (7 + 16)                    */
GPIOB->ODR   ^=  (1U << 7);     /* toggle PB7                          */
```

Work the same five lines out for LD2 on PA5 yourself: which clock bit, which MODER bits, which BSRR bits? `GPIOB` is a pointer to a `GPIO_TypeDef` structure, so `GPIOB->MODER` is a member reached through a pointer.

L05 — HAL GPIO

```
GPIO_InitTypeDef cfg = {0};
cfg.Pin  = GPIO_PIN_5;              /* a mask: 1U << 5                    */
cfg.Mode = GPIO_MODE_OUTPUT_PP;     /* MODER field 01, OTYPER bit 0       */
cfg.Pull = GPIO_NOPULL;             /* PUPDR field 00                     */
HAL_GPIO_Init(GPIOA, &cfg);         /* &cfg: the function reads your struct */
```

HAL name	Register effect for pin n
`GPIO_MODE_INPUT`	MODER field `00`
`GPIO_MODE_OUTPUT_PP`	MODER field `01`, OTYPER bit n `0`
`GPIO_MODE_IT_FALLING`	MODER `00` + EXTI falling edge + line unmasked
`GPIO_NOPULL` / `GPIO_PULLUP` / `GPIO_PULLDOWN`	PUPDR field `00` / `01` / `10`

- Run-time: `HAL_GPIO_WritePin(port, pin, GPIO_PIN_SET/RESET)`, `HAL_GPIO_TogglePin(port, pin)`, `HAL_GPIO_ReadPin(port, pin)` returns `GPIO_PIN_SET` (high) or `GPIO_PIN_RESET` (low).
- A **floating** input reads noise. A **pull-up** makes the released level high; a **pull-down** makes it low. If the board already has an external resistor, use `GPIO_NOPULL`.
- **Read the wiring, not the word "pressed":** a button that connects the pin to GND gives a *low* level when pressed; a button that connects the pin to 3.3 V gives a *high* level. Work out what `HAL_GPIO_ReadPin` returns from that.
- A button pressed for a long time is one press: detect the **edge** (previous level high, current level low), not the level.

L06 — Interrupts

```
press -> PC13 falls -> EXTI13 edge detect -> EXTI->PR bit 13 = 1 (pending)
      -> EXTI->IMR bit 13 must be 1 -> EXTI15_10_IRQn -> NVIC enable must be 1
      -> CPU saves its place -> vector table entry -> EXTI15_10_IRQHandler()   (processor enters)
      -> HAL_GPIO_EXTI_IRQHandler()  (clears PR)  -> HAL_GPIO_EXTI_Callback(pin)  (you write this)
```

- Two enable switches: `EXTI->IMR` (written by `HAL_GPIO_Init`) and the NVIC (`HAL_NVIC_EnableIRQ`). With the NVIC switch open, EXTI still sets the pending bit, but the handler never runs.

- `EXTI->PR` is **write-1-to-clear**; the HAL helper clears it. Your `button_event` variable is a different flag: `main()` clears it.
- **Callback rules:** check the pin, record the event, return. No `HAL_Delay()`, no loops, no LED code.
- A variable shared with an interrupt is **`volatile`**: otherwise the compiler may keep a stale copy in a register and `main()` never sees the change.
- **Debounce:** accept an edge only if `now - last >= GUARD` (unsigned subtraction, wrap-safe). Update `last` only on an accepted edge. `>=` accepts an edge exactly `GUARD` ms after the last one.

L07 — Timers

$$f_{\text{counter}} = \frac{f_{\text{timer clock}}}{\text{PSC} + 1} \qquad T_{\text{update}} = \frac{\text{ARR} + 1}{f_{\text{counter}}} \qquad f_{\text{update}} = \frac{f_{\text{timer clock}}}{(\text{PSC} + 1)(\text{ARR} + 1)}$$

- **Both +1s.** `PSC = 0` divides by 1. `ARR = 99` means 100 counts (0 … 99).
- Many `PSC`/`ARR` pairs give the same period; a smaller `PSC` gives finer resolution.
- `HAL_TIM_Base_Start(&htim)` runs the counter only. `HAL_TIM_Base_Start_IT(&htim)` also enables the update interrupt (`UIE`). Without `_IT` the callback never runs.
- Polling: **test** `TIMx->SR & TIM_SR_UIF`, then **clear** it. Forget the clear and the loop fires continuously.
- Interrupt: the HAL clears `UIF` and calls `HAL_TIM_PeriodElapsedCallback(htim)`. One callback serves all timers: before acting, compare the handle's timer-peripheral member with the timer you expect (the member name is on the exam's reference page).
- Tick pattern: the callback only adds one to the millisecond counter. `main()` schedules with `if ((g_ms - t_task) >= INTERVAL) { t_task = g_ms; /* work */ }`. No `HAL_Delay()` anywhere in that loop.

Practice set

Work on paper. Cover the answers. Part A practises Section A, Part B Section B, Part C Section C and Part D Section D. There are more practice items than exam questions: practise all of them.

Part A — Trace the execution

A1. What is printed?

```c
#include <stdio.h>
#include <stdint.h>

int main(void)
{
    uint16_t readings[5] = {1200, 4300, 3000, 1500, 3000};
    uint32_t sum = 0;
    int high = 0;

    for (int i = 0; i < 5; i++) {
        if (readings[i] > 4095) {
            readings[i] = 4095;
        }
        sum += readings[i];
        if (readings[i] >= 3000) {
            high++;
        }
    }
    printf("%u %d %u\n", sum, high, sum / 5);
    return 0;
}
```

A2. What is printed? Why does the first line not change?

```c
#include <stdio.h>

void double_it(int n)  { n = n * 2; }
int  twice(int n)      { return n * 2; }

int main(void)
{
    int v = 7;
    double_it(v);
    printf("%d\n", v);
    v = twice(v);
    printf("%d\n", v);
    return 0;
}
```

A3. What is printed?

```c
#include <stdio.h>

struct Reading { int id; int value; };

int main(void)
{
    int m = 10;
    int *p = &m;
    *p += 5;
    int n = *p - 3;

    struct Reading r = {1, 50};
    struct Reading *pr = &r;
    pr->value = n;
    r.id = m;
```

```
    printf("%d %d %d %d\n", m, n, r.id, pr->value);
    return 0;
}
```

A4. What is printed (2 hex digits)? If `r` were the **low byte** of `GPIOB->MODER` (pins 3…0; 00 = input, 01 = output, 10 = alternate, 11 = analog), which pin do the last two lines configure and to which mode?

```
#include <stdio.h>
#include <stdint.h>

int main(void)
{
    uint8_t r = 0xF0U;            /* binary 1111 0000 */
    r |=  (1U << 1);
    r &= ~(1U << 4);
    r &= ~(3U << 6);
    r |=  (2U << 6);
    printf("%02X\n", r);
    return 0;
}
```

A5. The guard is 50 ms. Falling edges call the callback at `HAL_GetTick()` = 300, 310, 345, 360, 420, 470 ms. Which edges are accepted, and what is `last` at the end?

```
uint32_t last = 0;
volatile uint8_t event = 0;

void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    uint32_t now = HAL_GetTick();
    if (GPIO_Pin == GPIO_PIN_13 && (now - last) >= 50) {
        last = now;
        event = 1;
    }
}
```

A6. `for (int i = 0; i <= 4; i++) printf("%d ", i * i);` — what is printed, and how many times does the body run?

Part B — Concept questions (one answer each)

B1. The build stops at `error: 'mean' undeclared`. Which stage reported it, and what is missing?

B2. The build stops at `undefined reference to 'read_sensor'`. Which stage reported it, and why did compilation succeed?

B3. A pin is configured as input with `GPIO_NOPULL` and nothing is connected to it. What does `HAL_GPIO_ReadPin` return?

B4. An external button connects PA0 to **3.3 V** when pressed, and nothing else is connected. Which pull setting do you choose, and what does a press read?

B5. `HAL_GPIO_Init(GPIOB, &cfg)` ran with `Pin = GPIO_PIN_7`, `Mode = GPIO_MODE_OUTPUT_PP`, `Pull = GPIO_PULLDOWN`. Which bits of MODER and PUPDR changed, and to what?

B6. Which function does the **processor** enter when EXTI line 13 fires, and which function does the **HAL** call for you?

B7. After a press, you pause the program inside the handler. What does `EXTI->PR` bit 13 read, and who will clear it?

B8. The timer clock is 16 MHz, `PSC = 7`, `ARR = 19999`. What is the counter frequency and the update period?

B9. The counter is configured correctly and `HAL_TIM_Base_Start(&htim2)` is called. `TIM2->CNT` changes every time you pause, but the callback never runs. Which bit is 0, and which one call fixes it?

B10. A polling loop tests `UIF` and toggles the LED, and the LED blinks far too fast. Which line is missing?

B11. Why must `g_ms` be declared `volatile`?

B12. `g_ms` is `uint32_t`. Why is `(g_ms - t) >= 500` still correct after `g_ms` wraps from `0xFFFFFFFF` to `0`?

Part C — Fill in the blank

C1. An external LED is on **PB0**. GPIOB's clock enable is `RCC->AHB1ENR` bit 1.

```
void led2_init(void)
{
    RCC->AHB1ENR |= (________);      /* GPIOB clock                  */
    GPIOB->MODER &= ~(________);     /* clear the PB0 mode field     */
    GPIOB->MODER |=  (________);     /* 01 = output                  */
}
void led2_on(void)  { GPIOB->BSRR = (________); }   /* PB0 high */
void led2_off(void) { GPIOB->BSRR = (________); }   /* PB0 low  */
```

C2. Count how many samples are above a limit.

```
int count_above(uint16_t values[], int count, uint16_t limit)
{
    int n = ________;
    for (int i = 0; i < ________; i++) {
        if (values[i] ________ limit) {
            n________;
        }
    }
    return ________;
}
```

C3. B1 on PC13, falling-edge EXTI. Count accepted presses in the callback; `main()` prints every tenth.

```
________ uint32_t press_count = 0;

void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if (GPIO_Pin == ________) {
        press_count________;
    }
}

/* in main() */
uint32_t shown = 0;
while (1) {
    if (press_count - shown >= ________) {
        shown = press_count;
        printf("%lu presses\n", (unsigned long)________);
    }
}
```

C4. Timer clock 16 MHz. Goal: counter frequency 20 kHz, one update event every 25 ms, by interrupt.

```
htim3.Instance       = TIM3;
htim3.Init.Prescaler = ________;     /* 16 MHz / (PSC + 1) = 20 kHz */
htim3.Init.Period    = ________;     /* 20 kHz / (ARR + 1) = 40 Hz  */
HAL_TIM_Base_Init(&htim3);
________(&htim3);

void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
    if (htim->Instance == ________) {
        HAL_GPIO_TogglePin(GPIOA, ________);     /* LD2 */
    }
}
```

C5. A button on **PC6** connects the pin to GND when pressed and has no external resistor. GPIOC's clock enable is `RCC->AHB1ENR` bit 2. In MODER, `00` means input; in PUPDR, `01` means pull-up.

```c
void button_init(void)
{
    RCC->AHB1ENR |= (________);      /* GPIOC clock                 */
    GPIOC->MODER &= ~(________);     /* 00 = input: clear the field */
    GPIOC->PUPDR &= ~(________);     /* clear the PC6 pull field    */
    GPIOC->PUPDR |=  (________);     /* 01 = pull-up                */
}
int button_pressed(void)
{
    return (GPIOC->IDR & (________)) == 0U;   /* pressed = pin low */
}
```

C6. **PA9** becomes a UART transmit pin, so its MODER field gets `10` (alternate function). An LED on **PA8** is toggled through `ODR` and switched off through `BSRR`. Pins above 7 work the same way: pin n owns MODER bits $2n{+}1{:}2n$.

```c
GPIOA->MODER &= ~(________);     /* clear the PA9 field              */
GPIOA->MODER |=  (________);     /* write 10: alternate function     */
GPIOA->ODR ____ (________);      /* toggle PA8 (XOR), other pins kept */
GPIOA->BSRR = (________);        /* PA8 low                          */
```

C7. Timer clock 16 MHz. Goal: counter frequency 10 kHz, one update event every 0.5 s, **no interrupt**: `main()` polls the update flag.

```c
htim3.Instance       = TIM3;
htim3.Init.Prescaler = ________;     /* 16 MHz / (PSC + 1) = 10 kHz */
htim3.Init.Period    = ________;     /* 10 kHz / (ARR + 1) = 2 Hz   */
HAL_TIM_Base_Init(&htim3);
________(&htim3);                    /* start the counter only      */

while (1) {
    if (TIM3->SR & ________) {       /* update flag set?            */
        TIM3->SR &= ~TIM_SR_UIF;     /* clear it                    */
        HAL_GPIO_TogglePin(GPIOB, GPIO_PIN_0);
    }
}
/* The PB0 LED changes state ____ times per second. */
```

C8. Two buttons with EXTI interrupts: START on PA0 and STOP on PA1. The callback only records which button was pressed; `main()` acts on it.

```c
________ uint8_t start_event = 0;
________ uint8_t stop_event  = 0;

void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if (GPIO_Pin == GPIO_PIN_0) {
        start_event = 1;
    } else if (GPIO_Pin == ________) {
        stop_event = 1;
    }
}

while (1) {
    if (start_event == 1) {
        start_event = ________;      /* consume the event */
        motor_on();
    }
    if (stop_event == 1) {
        stop_event = 0;
        ________;                    /* switch the motor off */
    }
}
```

Part D — Write the code

D1. Write `uint16_t minimum_sample(uint16_t values[], int count)` returning the smallest element; return `0U` when `count` is 0.

D2. Write `int text_length(char text[])` returning the number of characters before `'\0'`. Then say how many bytes `"STOP"` occupies as a C string.

D3. Write `void gpio_set_pull(GPIO_TypeDef *port, uint32_t pin, uint32_t pull)` that writes the 2-bit `pull` pattern (`00` none, `01` up, `10` down) into the PUPDR field of `pin` without changing other pins. The field of `pin` starts at bit $2 \cdot$ `pin`, so its mask is `3U << (2U * pin)`: clear it first, then OR the shifted pattern in. Then write `uint32_t gpio_read(GPIO_TypeDef *port, uint32_t pin)` that returns 1 when the pin is high and 0 when low, using `IDR`.

D4. With HAL, configure PB0 as a push-pull output (external LED) and PA0 as an input for a button that connects PA0 to 3.3 V when pressed (no external resistor). Fill a `GPIO_InitTypeDef` for each and call `HAL_GPIO_Init`.

D5. PC13 is `GPIO_MODE_IT_FALLING` with its NVIC line enabled. Write the callback (record a PC13 falling edge in a `volatile` flag; no debouncing needed, the callback must not wait or touch the LED) and the `while (1)` body that toggles LD2 on **every third** press. No polling, no `HAL_Delay()`.

D6. `g_ms` is the 1 ms tick counter. The loop below already toggles LD2 every **250 ms** without `HAL_Delay()`:

```
uint32_t t_led = 0;

while (1) {
    if ((g_ms - t_led) >= 250U) {
        t_led = g_ms;
        HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
    }
    /* your code */
}
```

Add a second task with the same pattern so that `log_values()` runs every **1 s**, still with no `HAL_Delay()`. Then explain in one sentence what changes if you write `t_led += 250U` instead of `t_led = g_ms`, and why `g_ms` must be `volatile`.

D7. Without interrupts: write a `while (1)` body that toggles LD2 on every TIM2 update event by **polling** `TIM2->SR` (`TIM_SR_UIF` is bit 0).

D8. Write `int first_over(uint16_t values[], int count, uint16_t limit)` that returns the index of the **first** element greater than `limit`, or `-1` when there is none. `first_over(values, 6, 3000)` with `{1200, 2900, 3100, 3500, 800, 3100}` returns 2.

D9. Write two more port-pointer helpers. `void gpio_toggle(GPIO_TypeDef *port, uint32_t pin)` inverts the pin through `ODR` without changing other pins. `void gpio_set_open_drain(GPIO_TypeDef *port, uint32_t pin, uint32_t on)` writes the **1-bit** `OTYPER` field of `pin` (`1` = open-drain, `0` = push-pull): clear the bit first, then OR the new value in.

D10. Two buttons, two LEDs. B1 (PC13) toggles LD2 (PA5); an external button on PA0 toggles an LED on PB0. Both pins are `GPIO_MODE_IT_FALLING` with their NVIC lines enabled. Write the callback (one `volatile` flag per button) and the `while (1)` body. No polling, no `HAL_Delay()`.

D11. A press of B1 turns LD2 **on**, and LD2 turns **off** by itself 2 s after the **last** press (a new press restarts the 2 s). `g_ms` is the 1 ms tick counter and the callback already sets `volatile uint8_t button_event = 1` on each press. Write the declarations and the `while (1)` body, with no `HAL_Delay()`.

Answers

Part A

A1. `12795 3 2559`. The 4300 is clamped to 4095 before it is added: 1200 + 4095 + 3000 + 1500 + 3000 = 12795. `>= 3000` counts 4095, 3000 and 3000 (three). 12795 / 5 = 2559 exactly.

A2. `7` then `14`. `double_it` changes its own copy `n`; `v` in `main()` is untouched. `twice` returns 14 and the caller stores it with `v = twice(v)`.

A3. `15 12 15 12`. `*p += 5` writes through the pointer, so `m` is 15; `n` is 12; `pr->value = n` sets `r.value`; `r.id = m`.

A4. `A2`. `0xF0 = 1111 0000` → set bit 1 → `1111 0010` (`0xF2`) → clear bit 4 → `1110 0010` (`0xE2`) → clear bits 7:6 → `0010 0010` (`0x22`) → write `10` at bits 7:6 → `1010 0010` (`0xA2`). Bits 7:6 belong to pin 3 (2 × 3 = 6): **PB3, alternate function** (`10`).

A5. Accepted: **300** (300 − 0 50), **360** (360 − 300 = 60), **420** (60), **470** (470 − 420 = 50, and `>=` accepts exactly 50). Rejected: 310 (10) and 345 (45). `last` ends at **470**. Four events.

A6. `0 1 4 9 16` — the body runs **5** times because the condition is `<=`.

Part B

B1. The **compiler**. A declaration of `mean` is missing before its first use (a variable declaration or a function prototype / header include).

B2. The **linker**. The header gave the compiler a declaration, so every call compiled. No object file in the build contains the definition: the `.c` file is not in the build list or the body was never written.

B3. A random, unstable value: the input is floating. Nothing defines the level.

B4. `GPIO_PULLDOWN`, so the released pin reads low. A press connects it to 3.3 V and reads `GPIO_PIN_SET` (active-high).

B5. Pin 7 owns bits 15:14. `MODER[15:14] = 01` (output), `PUPDR[15:14] = 10` (pull-down). Also `OTYPER` bit 7 = 0 (push-pull).

B6. The processor enters `EXTI15_10_IRQHandler()`, the function named in the vector table. The HAL then calls your callback, `HAL_GPIO_EXTI_Callback()`, with `GPIO_PIN_13` as its argument.

B7. It reads **1**: the edge is pending and not yet serviced. The HAL helper `HAL_GPIO_EXTI_IRQHandler()` clears it by writing 1 to the bit, before it calls your callback.

B8. Counter frequency 16 MHz / (7 + 1) = **2 MHz**. Update period (19999 + 1) / 2 MHz = 20000 × 0.5 µs = **10 ms** (100 Hz).

B9. `TIM2->DIER` bit 0 (`UIE`) is 0: the update event never raises an interrupt. Call `HAL_TIM_Base_Start_IT(&htim2)` instead (and keep the NVIC line enabled).

B10. The clear: `TIM2->SR &= ~TIM_SR_UIF;` (or `__HAL_TIM_CLEAR_FLAG(&htim2, TIM_FLAG_UPDATE)`). The flag stays set, so every pass of the loop "sees" an update.

B11. The interrupt changes `g_ms` between two instructions of `main()`. Without `volatile` the compiler may keep a copy in a register and reuse the stale value; `volatile` forces a fresh read from memory every time.

B12. Unsigned subtraction wraps the same way the counter does: for example `3 - 0xFFFFFFFE` gives 5, the true elapsed time. `g_ms >= t + 500` would not survive the wrap.

Part C

C1. `1U << 1`, `3U << 0` (or `3U`), `1U << 0` (or `1U`), `1U << 0`, `1U << 16`.

C2. `0`, `count`, `>`, `++`, `n`.

C3. `volatile`, `GPIO_PIN_13`, `++`, `10`, `press_count` (or `shown`).

C4. `799` (16 MHz / 800 = 20 kHz), `499` (20 kHz / 500 = 40 Hz = 25 ms), `HAL_TIM_Base_Start_IT`, `TIM3`, `GPIO_PIN_5`.

C5. `1U << 2`, `3U << 12`, `3U << 12`, `1U << 12`, `1U << 6`. PC6 owns bits 13:12 in MODER and in PUPDR (2 × 6 = 12); `IDR` has one bit per pin, so the read mask is `1U << 6`.

C6. `3U << 18`, `2U << 18`, `^=`, `1U << 8`, `1U << 24`. PA9 owns bits 19:18 (2 × 9 = 18), and `10` shifted there is `2U << 18`. In BSRR, bit 8 + 16 = 24 drives PA8 low.

C7. `1599` (16 MHz / 1600 = 10 kHz), `4999` (10 kHz / 5000 = 2 Hz = 0.5 s), `HAL_TIM_Base_Start`, `TIM_SR_UIF`, `2` (one toggle per update event).

C8. `volatile`, `volatile`, `GPIO_PIN_1`, `0`, `motor_off()`. Both flags are written by the interrupt and read by `main()`, so both need `volatile`; `main()` clears a flag before acting on it, so a press that arrives during `motor_on()` is not lost.

Part D

D1.

```c
uint16_t minimum_sample(uint16_t values[], int count)
{
    if (count == 0) {
        return 0U;
    }
    uint16_t minimum = values[0];
    for (int i = 1; i < count; i++) {
        if (values[i] < minimum) {
            minimum = values[i];
        }
    }
    return minimum;
}
```

D2.

```c
int text_length(char text[])
{
    int i = 0;
    while (text[i] != '\0') {
        i++;
    }
    return i;
}
```

`"STOP"` occupies **5** bytes: four characters and the `'\0'` terminator.

D3.

```c
void gpio_set_pull(GPIO_TypeDef *port, uint32_t pin, uint32_t pull)
{
    port->PUPDR &= ~(3U << (2U * pin));           /* clear the field */
    port->PUPDR |=  ((pull & 3U) << (2U * pin));  /* write the pattern */
}

uint32_t gpio_read(GPIO_TypeDef *port, uint32_t pin)
{
    return (port->IDR >> pin) & 1U;
}
```

D4.

```c
GPIO_InitTypeDef cfg = {0};

cfg.Pin  = GPIO_PIN_0;              /* PB0: external LED        */
cfg.Mode = GPIO_MODE_OUTPUT_PP;
cfg.Pull = GPIO_NOPULL;
HAL_GPIO_Init(GPIOB, &cfg);

cfg.Pin  = GPIO_PIN_0;              /* PA0: button to 3.3 V     */
cfg.Mode = GPIO_MODE_INPUT;
cfg.Pull = GPIO_PULLDOWN;           /* released reads low       */
HAL_GPIO_Init(GPIOA, &cfg);
```

D5.

```c
volatile uint8_t button_event = 0;
uint8_t  presses = 0;

void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if (GPIO_Pin == GPIO_PIN_13) {
        button_event = 1;                /* record only */
    }
}

while (1) {
    if (button_event == 1) {
        button_event = 0;                /* consume first */
        presses++;
        if (presses == 3) {
            presses = 0;
            HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
        }
    }
}
```

The L06 lab's 50 ms guard (`uint32_t now = HAL_GetTick();` and `(now - last_press_ms) >= 50U` in the callback, with `last_press_ms = now;` on an accepted edge) is a good extra, but the exam does not ask for it in the code question; the trace question (A5) covers it.

D6.

```c
uint32_t t_log = 0;                  /* declared next to t_led */

    /* at the "your code" comment, a sibling of the LED block */
    if ((g_ms - t_log) >= 1000U) {
        t_log = g_ms;
        log_values();
    }
```

`t_led += 250U` keeps every deadline on the original 250 ms grid, so a late pass delays one toggle but not the following ones; `t_led = g_ms` re-anchors to the late moment and the lateness accumulates. `g_ms` is written by the TIM2 interrupt outside the flow of `main()`; `volatile` forces every read to go to memory, so the loop sees each new value instead of a copy the compiler kept in a register.

D7.

```c
while (1) {
    if (TIM2->SR & TIM_SR_UIF) {        /* 1. test the flag  */
        TIM2->SR &= ~TIM_SR_UIF;        /* 2. clear it       */
        HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
    }
}
```

The counter must be running (`HAL_TIM_Base_Start(&htim2)` or `TIM2->CR1 |= 1U`).

D8.

```c
int first_over(uint16_t values[], int count, uint16_t limit)
{
    for (int i = 0; i < count; i++) {
        if (values[i] > limit) {
            return i;                /* stop at the first match */
        }
    }
    return -1;                       /* after the loop: none found */
}
```

D9.

```c
void gpio_toggle(GPIO_TypeDef *port, uint32_t pin)
{
```

```
    port->ODR ^= (1U << pin);
}

void gpio_set_open_drain(GPIO_TypeDef *port, uint32_t pin, uint32_t on)
{
    port->OTYPER &= ~(1U << pin);              /* clear the bit   */
    port->OTYPER |=  ((on & 1U) << pin);       /* write 0 or 1    */
}
```

A 1-bit field uses the mask `1U << pin`; a 2-bit field (MODER, PUPDR) uses `3U << (2U * pin)`. The method is the same: clear first, then OR.

D10.

```
volatile uint8_t b1_event = 0;
volatile uint8_t ext_event = 0;

void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if (GPIO_Pin == GPIO_PIN_13) {
        b1_event = 1;
    } else if (GPIO_Pin == GPIO_PIN_0) {
        ext_event = 1;
    }
}

while (1) {
    if (b1_event == 1) {
        b1_event = 0;
        HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
    }
    if (ext_event == 1) {
        ext_event = 0;
        HAL_GPIO_TogglePin(GPIOB, GPIO_PIN_0);
    }
}
```

One callback serves every EXTI line, so it must look at `GPIO_Pin`.

D11.

```
uint32_t t_press = 0;
uint8_t  led_on  = 0;

while (1) {
    if (button_event == 1) {
        button_event = 0;
        t_press = g_ms;                         /* restart the 2 s */
        led_on = 1;
        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
    }
    if (led_on == 1 && (g_ms - t_press) >= 2000U) {
        led_on = 0;
        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);
    }
}
```

The same subtraction test as the scheduler, but the timestamp is set by an event (the press) instead of by the task itself.

If you can do every item here without looking at the answers, you are ready for the exam. If one lecture gives you trouble, go back to that lecture's study sheet and worksheet.